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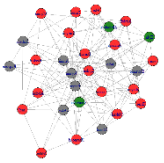

Team #: 3

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WORKSHOP: Network Analysis and Targeting with R – part II

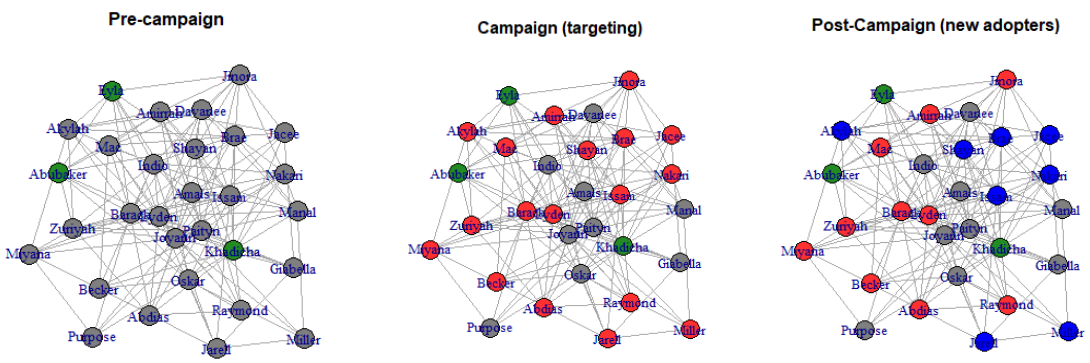
Objective: In this assignment you are helping a partnering firm to select targets for an upcoming promotion campaign. The promotion offers a deep discount for the company's product to stimulate product purchase (adoption). You are given access to the historical data collected in the previous similar campaign that includes consumers product adoption status prior (pre_adopter) and post (post_adopters) the campaign for all - targeted (first_target) and not targeted consumers. Additionally, you have information about social ties of all of your prospect targets:

	name	pre_adopter	first_target	post_adopter
1	Giabella	0	0	0
2	Raymond	0	1	0
3	Miyana	0	1	0
4	Purpose	0	0	0
5	Miller	0	1	1
6	Oskar	0	0	0
7	Eyla	1	0	0
8	Manal	0	0	0
9	Nakari	0	1	1
10	Abdias	0	1	0

	name_from	name_to
1	Giabella	Jacee
2	Giabella	Jarell
3	Giabella	Paityn
4	Giabella	Lyden
5	Giabella	Barack
6	Giabella	Joyann
7	Raymond	Oskar
8	Raymond	Abdias
9	Raymond	Paityn
10	Raymond	Shayan

Note that post_adopter column indicates the response to the campaign (if targeted) but not the adoption status prior to the first campaign (e.g. pre campaign adopters have a value of zero in this column).

The figures below depict 3 stages captured in the historical data:



a. b. c.

At the Pre-campaign stage (a) the company had a few adopters (marked in green) who purchased the product without a promotion. At the Campaign stage (b) the company selected some prospects (marked in red) *at random* and offered them a discount. At the Post-campaign stage (c) the company observed

who among the targeted individuals have purchased the product (marked in blue). Your task is to identify potential targets among the prospects (marked in grey at stage (c)) for the next campaign.

Your analysis will include the following steps:

1. Using historical data develop a **product adoption model** to identify factors predictive for target's response to promotion. Some tips for this part:
    a. use binary logit to model the adoption decision
    b. to form your predictors, use the information about pre-campaign adoption status and node's network attributes:
        i. Neighborhood size (i.e. degree)
        ii. Number of neighbors who own the product
        iii. Neighbors' connectivity (i.e. clustering or transitivity)
        iv. Community effects (as some communities might be more responsive to the promotion)
    c. To **calibrate** the model, use **only the observations** for consumers who were targeted in the first promotion
    d. After the model has been estimated, drop the predictors whose effects on adoption are not statistically significant and recalibrate the model
    e. Demonstrate model performance using cumulative gains chart

2. Use the model developed in 1.d to predict adoption probability of each of the current non-adopters not previously targeted with the promotion.
    A tip for this part:
    You need to update the number of neighbors-adopters before making prediction with your calibrated model to reflect the current (pre-second campaign) adoption status of neighbors of your prospects.

3. Identify the top 100 individuals who are most likely to adopt the product if targeted in the upcoming promotion. What is the expected number of new adopters if you target top 100 prospects?

Execution and Deliverables

1. Download the R code and two data files from the class website. The parts of R code that require your coding are marked with **# YOUR CODE HERE**

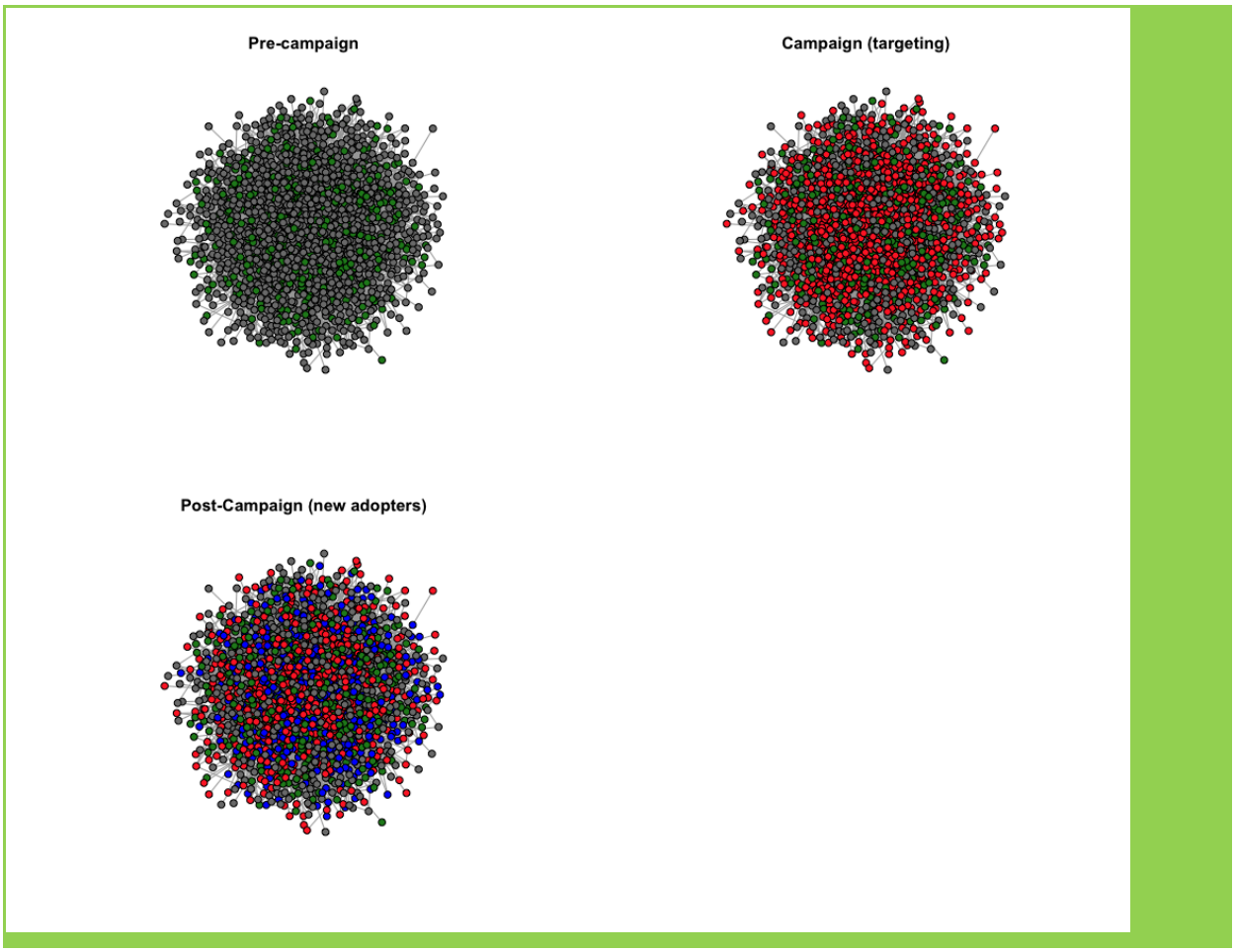
2. Load the R code in RStudio, set a breakpoint in load_network() function, run the code to the breakpoint and familiarize yourself with the input data – i.e. tables NetworkNodes and NetworkEdges. (The same approach can be used to get other results listed below)

3. Provide the following descriptives of your data (Hint: look under data_descriptives() function):
    a. Number of consumers: 1992
    b. Number of adopters prior to the first campaign: 401
    c. Number of consumers targeted in the first campaign: 971
    d. Number of new adopters generated through the first campaign: 396
    e. First campaign response rate (i.e. Adopted/Targeted): 0.407827
    f. Total number of adopters at this point (i.e. prior to the second campaign): 797
    g. Total number of potential targets (i.e. non-adopters not targeted before): 620

4. Provide the following descriptives for the network (Hint: look under network_descriptives() function):
    a. Number of connections in the network: 10000
    b. Average node degree: 10.04
    c. Degree distribution (histogram): Right-skewed; most users have low degree, few users have very high degree
    d. Clustering (transitivity) distribution (histogram): Mostly low clustering; most users' friends are not strongly connected
    e. Average (across all users) number of neighbors-adopters *before* the first campaign: 1.96
    f. Average (across all users) number of neighbors-adopters *after* the first campaign: 3.78
    g. Number of communities: 11
    h. Communities by size (bar chart): Largest communities: 8, 4, 9, and 7; smallest: 11 and 10

5. Provide the plot of the 3 stages reflected in historical data (use network_visualization() function)

YOUR PLOT HERE



6. For each user in the network generate attributes that will serve as an input to your model model (Hint: look under Create_node_level_attributes() function):
    i. Neighborhood size (i.e. degree)
    ii. Number of neighbors who own the product
    iii. Neighbors' connectivity (i.e. clustering or transitivity)
    iv. Community membership dummies

    Inspect node_level_data table to make sure you understand all the variables formed.

7. Prepare model input (review the Generate_model_input() function code making sure you follow the steps).

8. Build a response model based on the results of the first campaign. (Hint: look under Calibrate_Response_model() function):
    a. set up and estimate the full model

model_formula = "post_adopter ~ clustering + neighbors + neighborspreadopters + Com2 + Com3 + Com4 + Com5 + Com6 + Com7 + Com8 + Com9 + Com10 + Com11"

FULL MODEL ESTIMATION RESULTS HERE

```
[1] "Estimation Results - FULL model"
                        Estimate Std. Error    z value     Pr(>|z|)
(Intercept)          -1.07108877  0.3359249 -3.1884767 1.430245e-03
clustering            4.31406280  0.8712698  4.9514661 7.365644e-07
neighbors            -4.87919039  0.7901633 -6.1749142 6.619939e-10
neighborspreadopters  3.21655046  0.5613390  5.7301390 1.003484e-08
Com2                 -0.87045231  0.5214503 -1.6692909 9.505973e-02
Com3                  0.22333267  0.3596540  0.6209653 5.346225e-01
Com4                  3.18664868  0.4020323  7.9263508 2.256796e-15
Com5                  0.09141261  0.4958148  0.1843685 8.537244e-01
Com6                 -0.04209721  0.4218407 -0.0997941 9.205078e-01
Com7                  0.26987619  0.3498414  0.7714244 4.404554e-01
Com8                 -3.06258891  0.5867487 -5.2195918 1.793179e-07
Com9                  1.81473077  0.3566046  5.0889161 3.601159e-07
Com10                 1.21506676  0.5944572  2.0439937 4.095416e-02
Com11                -0.02899768  1.0530942 -0.0275357 9.780325e-01
[1] "Estimation Results - SELECTED model"
              Estimate Std. Error   z value     Pr(>|z|)
(Intercept) -0.3814398 0.06633873 -5.749881 8.930629e-09
clustering   2.1373409 0.70972199  3.011519 2.599446e-03
>
```

b. in the full model (from step a) identify the predictors that are statistically significant (Pr(>|z|) < .05) and drop other predictors from the model

model_formula = "post_adopter ~ clustering + neighbors + neighborspreadopters + Com8 + Com9 + Com10"

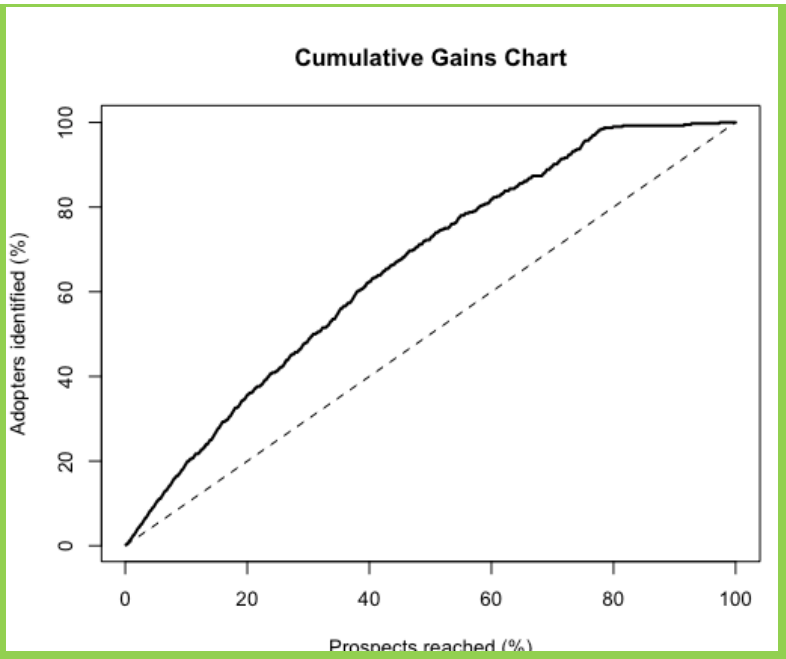
COMMENT ON WHAT PREDICTORS DID YOU DROP : Dropped Com2, Com3, Com4, Com5, Com6, Com7, and Com11 because they were not statistically significant at the 0.05 level.

c. re-estimate the model and create a gain chart based on historic data

SELECTED MODEL ESTIMATION RESULTS HERE

```
[1] "Estimation Results - SELECTED model"
                        Estimate Std. Error    z value     Pr(>|z|)
(Intercept)          -0.3447085  0.1484546 -2.3219789 2.023407e-02
clustering            2.8121774  0.7668676  3.6670963 2.453204e-04
neighbors            -2.6010366  0.6377815 -4.0782567 4.537466e-05
neighborspreadopters  2.4173602  0.4833315  5.0014538 5.689961e-07
Com8                 -3.7536859  0.5147186 -7.2926948 3.038157e-13
Com9                  0.8810874  0.1964041  4.4860944 7.254061e-06
Com10                 0.3507201  0.5044526  0.6952489 4.868993e-01
```

YOUR GAIN CHART PLOT HERE (A SAMPLE WITHOUT GAIN CURVE SHOWN BELOW)



Comment on the Gain Chart : The gains curve is above the random baseline, indicating that the selected model improves targeting efficiency compared with random selection. The model identifies a larger share of adopters by targeting a smaller share of prospects.

9. Identify targets for the second campaign (Hint: look under Generate_targeting_predictions() function)
    a. Predict response probability of all prospects
    b. List the top 10 targeting prospects that has the highest probability to adopt in the second campaign:

Rank	Name	Adoption probability
1	Joyann	0.9485773
2	Aitor	0.9473648
3	Sierrah	0.8918147
4	Brae	0.8862295
5	Armonii	0.8817609
6	Trevelle	0.8817609
7	Legennd	0.8817609
8	Sakura	0.8817609

9	Gabreil	0.8817609
10	Vyana	0.8802014

c. If you offer promotion to top 100 prospects identified by your model, how many new adopters you may expect?

Expected number of adopters = 81.12445

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